

# The state of open source in subsurface

This survey attempts to capture the activities and attitudes around open source software, open data, and open content in the field of subsurface science and engineering.

The survey will be open from 18 May 2026 until 1 June 2026. The results will be shared at the Open for Energy workshop at the 2026 EAGE Annual Conference in Aberdeen, UK, on 7 June 2026, and published in this GitHub repo: [open-source-subsurface-survey-2026](#)

Thank you for taking part in this anonymous survey!

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\* Indicates required question

## About you & your organization

First, some basic information to help understand your point of view.

'Organization' means 'where you work' — or whatever other affiliation makes most sense when thinking about your professional relationship to open source software.

### 1. In which region are you located? \*

*Mark only one oval.*

- ☐ North America
- ☐ South America
- ☐ Europe
- ☐ Africa / Middle East
- ☐ Asia
- ☐ Australasia

### 2. What is your primary, current technical domain? \*

*Mark only one oval.*

- ☐ Pure sciences: maths, physics, chemistry
- ☐ Applied sciences, eg economic geology, geophysics
- ☐ Well / field / site related (operations, drilling)
- ☐ Reservoir engineering, geomodeling
- ☐ Mechanical & process engineering
- ☐ Business & financial modeling
- ☐ Software engineering, computer science, data science
- ☐ Other:  
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### 3. During the last 6 months, how much of your work time have you spent writing or reviewing code or documentation? \*

*Mark only one oval.*

- ☐ I don't write or review code or documentation
- ☐ Less than 20% (one day per week)
- ☐ Between 20% and 50%
- ☐ More than 50%

4. Which sector best describes your organization? \*

Mark only one oval.

- ☐ Academia or research institute
- ☐ Government or regulatory body
- ☐ Operating company
- ☐ Service or technology company
- ☐ Independent consultant or freelance
- ☐ Other:  
\_\_\_\_\_

5. What size is your organization? \*

Mark only one oval.

- ☐ 1 to 9 employees
- ☐ 10 to 99 employees
- ☐ 100 to 999 employees
- ☐ 1000 to 9999 employees
- ☐ 10 000 or more employees

#### Your use of open-source software

6. When evaluating technical tools for use in a project, how heavily do you and/or your team weigh whether a software option was open-source vs. proprietary? \*

Mark only one oval.

- ☐ Strongly prefer proprietary options
- ☐ Lean toward proprietary options
- ☐ Neutral / No weight
- ☐ Lean toward open-source options
- ☐ Strongly prefer open-source options

7. In the past 6 months, how often have you **used** open-source software for subsurface or other technical processing, modeling, or analysis? \*

Mark only one oval.

- ☐ Never, or I don't perform this kind of work      *Skip to question 10*
- ☐ Rarely
- ☐ Monthly
- ☐ Weekly
- ☐ Daily
- ☐ I don't know if I use open-source software or not      *Skip to question 10*

#### Your use of open-source software

8. Thinking about the open-source subsurface toolset you used in the past 6 months, what were the main reasons for choosing it over a proprietary alternative? \*

*Check all that apply.*

- ☐ Cost / No licensing fees
- ☐ Transparency and mathematical auditability of the underlying physics/algorithms
- ☐ Extensibility (ability to write custom plugins/scripts)
- ☐ Better interoperability and alignment with open data standards
- ☐ It was required by a client, partner, or academic collaborator
- ☐ There are no proprietary alternatives

9. In the past 6 months, have you encountered a workflow-breaking bug or security vulnerability in a subsurface open-source tool that delayed your project? \*

*Mark only one oval.*

- ☐ Yes, multiple times
- ☐ Yes, once
- ☐ No
- ☐ Unsure

#### Your activities as a core contributor or maintainer

10. Are you a **core contributor** to any open source repositories? (In any domain, not only subsurface.) \*

A 'core contributor' is someone with administrative or write access to the repository. This includes the role of 'maintainer'.

*Mark only one oval.*

- ☐ No *Skip to question 16*
- ☐ Yes, one
- ☐ Yes, more than one

#### Your activities as a core contributor or maintainer

11. Of the repositories to which you are a core contributor, are any of the projects related to subsurface? \*

*Mark only one oval.*

- ☐ No
- ☐ Yes, one
- ☐ Yes, more than one
- ☐ Yes, all of them

12. Thinking about the primary subsurface open source project you contribute to, how many other core contributors are there on that project? \*

*Mark only one oval.*

- ☐ There are no other core contributors
- ☐ One other
- ☐ 2 to 5 others
- ☐ More than 5 others

13. Again thinking about this project, how was your time accounted for? \*
- Mark only one oval.
- ☐

Fully supported and done during my paid organizational working hours
- ☐

Partially supported (done occasionally/informally during working hours)
- ☐

Entirely unsupported (done on my own personal/volunteer time)
- ☐

Not applicable (I did not make contributions in the past 6 months)

14. Again thinking about this project, what was the primary funding mechanism keeping the subsurface open-source tools you maintain or help maintain afloat? \*
- Mark only one oval.
- ☐

A single industry source, eg private company
- ☐

Public academic grants or government-funded research initiatives
- ☐

Joint Industry Projects (JIPs) / Corporate consortiums
- ☐

Dedicated foundational funding (e.g., NumFOCUS, Linux Foundation)
- ☐

Purely unfunded, volunteer in-kind developer time
- ☐

Unknown or unsure

15. Looking ahead to the next 6 to 12 months as a core contributor or maintainer, how do you feel about the following aspects: \*
- Mark only one oval per row.

|                                | Anxious<br>or<br>exposed | Nervous               | So-so                 | Hopeful               | Excited<br>or safe    |
|--------------------------------|--------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Shipping features              | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Fixing bugs                    | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Reviewing contributions        | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Getting recognition            | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Contributors' use of AI        | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Your project's use of AI       | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Upstream supply chain attacks  | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Your project(s) being attacked | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Project administration         | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Community management           | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Funding the work               | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Strategic planning             | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

### Your activities as a non-core contributor

16. In the past 6 months, how many pull requests (PRs), code commits, or documentation fixes have you submitted to open-source repositories to which you are **not** a core contributor? \*

Mark only one oval.

- ☐ None      Skip to question 20
- ☐ 1 or 2
- ☐ 3 to 5
- ☐ 6 to 10
- ☐ More than 10

17. In the past 6 months, what have been the barriers to contributing code or documentation back to the subsurface tools you use? \*

Check all that apply.

- ☐ I did not know, or was not sure, that I could contribute
- ☐ Lack of institutional time, permission, or career incentives
- ☐ Overly complex contribution pipelines (e.g., confusing CLA sign-offs, rigid CI/CD testing)
- ☐ Lack of clear developer documentation or onboarding guidance
- ☐ Unresponsive or unwelcoming project maintainers
- ☐ No barriers, or the need did not arise

### Your activities as a non-core contributor

18. Thinking about your contributions in the last 6 months, what proportion of them were submitted to **subsurface** open-source repositories? \*

Mark only one oval.

- ☐ None of them
- ☐ A few of them
- ☐ Most of them
- ☐ All of them

19. In the past 6 months, what kind of contribution(s) have you made? Check all that apply. \*

Check all that apply.

- ☐ Created issues or enhancement requests
- ☐ Typos and small fixes to documentation
- ☐ More substantial fixes to documentation or comments
- ☐ Non-critical bug fixes
- ☐ Critical bug fixes
- ☐ Features
- ☐ Improved existing tests
- ☐ Added new tests
- ☐ Data contributions
- ☐ Structured user testing and feedback
- ☐ Security-related input
- ☐ CI/CD, infrastructure or devops support
- ☐ Administrative support, fundraising, or project steering

### Artificial intelligence

For some, AI assistants like ChatGPT and agentic coding tools like Claude Code are affecting every part of the software development process. We'd like to understand how you are experiencing these changes.

20. How often do you use AI assistants or agentic tools in your work? For example, for help with research, brainstorming, translation, or coding.

Mark only one oval.

- ☐ Continuously
- ☐ Daily
- ☐ Weekly
- ☐ Rarely
- ☐ Never

21. To what extent do you trust the assistance of AI assistants and agentic tools? \*

Mark only one oval.

12345678910

Not☐☐☐☐☐☐☐☐☐☐ Completely

22. In the past 6 months, have you used AI code assistants (e.g., Copilot, ChatGPT, Claude) to write, debug, or document code meant for subsurface open-source applications? \*

Mark only one oval.

- ☐ Yes, frequently
- ☐ Yes, occasionally
- ☐ No, I code but I don't use AI for programming tasks
- ☐ No, I don't code

23. How concerned are you about the security, copyright, or intellectual property (IP) risks of AI-assisted code being merged into open-source repositories your organization **depends on**? \*

Mark only one oval.

12345

Not☐☐☐☐☐ Very concerned

24. How concerned are you about the security, copyright, or intellectual property (IP) risks of AI-assisted code being merged into open-source repositories your organization **produces**? \*

Mark only one oval.

12345

Not☐☐☐☐☐ Very concerned

25. To what extent do you agree with this statement: "AI code assistants are lowering the barrier for non-programmers to **make use of** subsurface open source projects." \*

Mark only one oval.

12345

Stro☐☐☐☐☐ Strongly agree

26. To what extent do you agree with this statement: "AI code assistants are lowering the barrier for non-programmers to **contribute to** subsurface open source projects."

Mark only one oval.

|      |                       |                       |                       |                       |                       |
|------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|      | 1                     | 2                     | 3                     | 4                     | 5                     |
| Stro | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
|      | Strongly agree        |                       |                       |                       |                       |

Funding and sustainability

27. In the past 6 months, how would you characterize your organizational leadership's general attitude toward adopting open-source software? \*

Mark only one oval.

☐ Enthusiastic (Viewed as a core competitive advantage)

☐ Supportive but passive (Permitted, but not actively strategized)

☐ Curious

☐ Indifferent or unaware

☐ Skeptical / Risk-averse (Preferred proprietary alternatives due to liability concerns)

☐ Actively hostile (Explicitly banned or heavily discouraged)

28. Which of the following actions has your organization taken? Select all that apply. \*

Check all that apply.

☐ Expressed a preference for open source software

☐ Defined a clear and visible open source strategy

☐ Joined or associated with open source organizations

☐ Funded open source (via foundation membership, sponsorship of individual developers, donations, etc)

☐ Explicitly provided engineering or infrastructure support to open source projects

☐ Implemented an Open Source Program Office (OSPO) or similar open source team

☐ Defined a public position on open source

☐ None of the above

29. In the last 6 months, and to the best of your knowledge, have any of your projects been affected by an upstream dependency doing any of the following: \*

Mark only one oval per row.

|                            |                       |                                              |                                              |                                    |
|----------------------------|-----------------------|----------------------------------------------|----------------------------------------------|------------------------------------|
|                            | No                    | Yes,<br>with<br>negative<br>effects<br>on us | Yes,<br>with<br>positive<br>effects<br>on us | Yes,<br>with no<br>effect<br>on us |
| Changing its license       | <input type="radio"/> | <input type="radio"/>                        | <input type="radio"/>                        | <input type="radio"/>              |
| Being forked               | <input type="radio"/> | <input type="radio"/>                        | <input type="radio"/>                        | <input type="radio"/>              |
| Being deprecated           | <input type="radio"/> | <input type="radio"/>                        | <input type="radio"/>                        | <input type="radio"/>              |
| Being attacked or infected | <input type="radio"/> | <input type="radio"/>                        | <input type="radio"/>                        | <input type="radio"/>              |

30. Based on your observations in the past 6 months, how sustainable do you consider the maintenance health of the subsurface open-source ecosystem?

Mark only one oval.

- ☐ Highly sustainable (Well-funded, robust contributor pipelines)
- ☐ Moderately sustainable (Stable but reliant on a few key individuals/companies)
- ☐ Fragile / At risk of severe maintainer burnout or code rot
- ☐ Critical state (Several tools we use appear abandoned)

31. That's the end of the survey. Is there anything you'd like to share?

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